

# title

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here is the sentence on the title page, you can set it with ``title_setence`` parameter.

# 1. Introduction

This is the Typst version of the original LaTeX `template_section.tex`.

中文测试 the machine has too much delay, i need to replace it with a more powerful one.  
为什么是楷体  
whatever, it is better than latex's default font. infinite times.

## section\_level = section

In this mode, section is the highest level and there is usually only one main section. There is no full title page and no cover image. All theorem-like environments share one counter, with numbering continuous through the document.

**Definition 1:** Let  $G$  be a group. A subgroup  $H \subset G$  is normal if  $gHg^{-1} = H$  for every  $g \in G$ .

**Theorem 2** (a great man): If  $H$  is a subgroup of index 2 in  $G$ , then  $H$  is normal.

*Proof:*

**Step 1:** The left cosets of  $H$  partition  $G$  into exactly two parts.

**Step 2:** Every left coset is also a right coset because there are only two cosets. Test the typst, it can be

**Step 3:** Therefore  $gH = Hg$  for all  $g \in G$ , so  $H$  is normal.

Hence we finish the proof of [?]. □

**Remark 3** (a great man): You can switch to `section_level: "chapter"` or `section_level: "book"` in `#show: article_settings.with(...)` to enable a cover page behavior similar to the class file. Here i test the ref to the theorem, it is [?].

Here we test the ref to step again, it is [?]. and the ref to the step in the second proof is [?].

**Proposition 4:** The intersection of two normal subgroups is normal.

*Solution:* let  $H_1$  and  $H_2$  be normal subgroups of  $G$ . We want to show that  $H_1H_2$  is normal.

**Step 1:** Let  $g \in G$  and  $h \in H_1H_2$ . Since  $H_1$  is normal, we have  $ghg^{-1} \in H_1$ . Similarly, since  $H_2$  is normal, we have  $ghg^{-1} \in H_2$ .

**Step 2:** Therefore,  $ghg^{-1} \in H_1H_2$ , proving that  $H_1H_2$  is normal. □

**Slogan:** “Mathematics is not about numbers, equations, computations, or algorithms: it is about understanding.” - William Paul Thurston

Test command  $\mathbb{A} \times \mathcal{T} \otimes \mathcal{I} \mathcal{J} f \mathbb{k} \hbar om \mathcal{M} \hbar om 1 \mathcal{M} \mathbb{I} \mathcal{J}_i \mathbb{I} \mathcal{T}_M$

Test the ref to the slogan, it is [?].

**Lemma 5:** Let  $G$  be a group and let  $H$  be a subgroup of  $G$ . If  $H$  is normal in  $G$ , then the quotient group  $G/H$  is well-defined.

*Proof:* Since  $H$  is normal in  $G$ , we can define the set of left cosets of  $H$  in  $G$ , denoted by  $G/H$ . The elements of  $G/H$  are of the form  $gH$  for some  $g \in G$ . We can define a binary operation on  $G/H$  by  $(gH)(hH) = (gh)H$ . This operation is well-defined because if  $g' \in gH$  and  $h' \in hH$ , then  $g' = gh_1$  and  $h' = hh_2$  for some  $h_1, h_2 \in H$ . Therefore,  $(g'H)(h'H) = (gh_1)(hh_2)H = (gh)(h_1h_2)H = (gh)H$ , which shows that the operation is independent of the choice of representatives. With this operation,  $G/H$  forms a group, called the quotient group of  $G$  by  $H$ . Thus, we have shown that if  $H$  is normal in  $G$ , then the quotient group  $G/H$  is well-defined.

**Case 1:** If  $H$  is the trivial subgroup  $\{e\}$ , then  $G/H$  is isomorphic to  $G$  itself, which is a group.

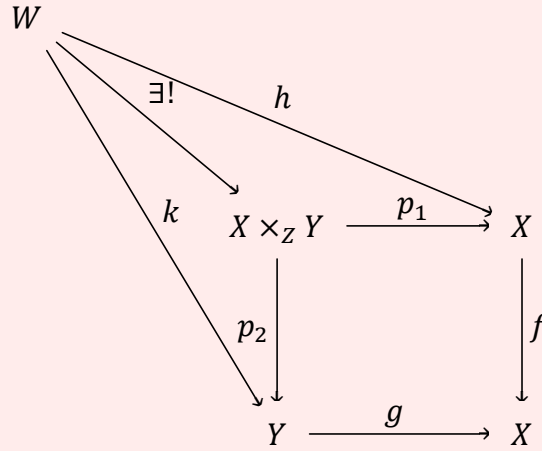
**Case 2:** If  $H$  is the whole group  $G$ , then  $G/H$  is the trivial group  $\{e\}$ , which is a group.

We have covered all cases, so the proof is complete.  $\square$

Here we test the ref to the case, it is [?] 2.

Here we test a ref to nothing, it is [?].

**Theorem 6:** In this theorem, we draw the Castersion diagram of the fiber product of two morphisms  $f : X \rightarrow Z$  and  $g : Y \rightarrow Z$  in the category of schemes. The diagram is as follows:



Then I can say that  $f : X \rightarrow Y$ .

Here is the reference. Let me test [?] and [?].

# Why I do not use Typst now

1. the syntactic sugar “/” can not be banned. For mathematical documents, we use “/” to denote the quotient (group, space, etc.) and it is very common while the fraction line is not suitable for this purpose.
2. the indent after a display math is not supported, which is very annoying for a mathematical document.

3. it can't define a new counter for the theorem-like environments
4. it has no support for the symbol “`\varprolim`”.

# Bibliography